SG No. 134 D_{4h}^{12} $P4_2/nnm$ [tetragonal]

* plus set: +[0, 0, 0]

Table 1: Wyckoff site: `2a`, site symmetry: `-42m`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[1,2,5,6,11,12,15,16]`
2	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[3,4,7,8,9,10,13,14]`

Table 2: Wyckoff site: `2b`, site symmetry: `-42m`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[1,2,5,6,11,12,15,16]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[3,4,7,8,9,10,13,14]`

Table 3: Wyckoff site: `4c`, site symmetry: `222.`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[1,2,5,6]`
2	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[3,4,7,8]`
3	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[9,10,13,14]`
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[11,12,15,16]`

Table 4: Wyckoff site: `4d`, site symmetry: `2.22`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[1,2,7,8]`
2	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[3,4,5,6]`
3	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[9,10,15,16]`
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[11,12,13,14]`

Table 5: Wyckoff site: `4e`, site symmetry: `..2/m`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,7,9,15]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[2,8,10,16]`
3	$\left[\frac{1}{2}, 0, 0\right]$	`[3,5,11,13]`
4	$\left[0, \frac{1}{2}, 0\right]$	`[4,6,12,14]`

Table 6: Wyckoff site: `4f`, site symmetry: `..2/m`

No.	position	mapping
1	[0, 0, 0]	`[1,7,9,15]`
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[2,8,10,16]`

continued ...

Table 6

No.	position	mapping
3	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[3,5,11,13]`
4	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[4,6,12,14]`

Table 7: Wyckoff site: `4g`, site symmetry: `2.mm`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[1,2,15,16]`
2	$\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[3,4,13,14]`
3	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[5,6,11,12]`
4	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	`[7,8,9,10]`

Table 8: Wyckoff site: `8h`, site symmetry: `2..`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	`[1,2]`
2	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	`[3,4]`
3	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[5,6]`
4	$\left[\frac{1}{4}, \frac{1}{4}, -z\right]$	`[7,8]`
5	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	`[9,10]`
6	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	`[11,12]`
7	$\left[\frac{3}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[13,14]`
8	$\left[\frac{3}{4}, \frac{3}{4}, z\right]$	`[15,16]`

Table 9: Wyckoff site: `8i`, site symmetry: `.2.`

No.	position	mapping
1	$\left[x, \frac{1}{4}, \frac{3}{4}\right]$	`[1,6]`
2	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{3}{4}\right]$	`[2,5]`
3	$\left[\frac{1}{4}, x, \frac{1}{4}\right]$	`[3,7]`
4	$\left[\frac{1}{4}, \frac{1}{2}-x, \frac{1}{4}\right]$	`[4,8]`
5	$\left[-x, \frac{3}{4}, \frac{1}{4}\right]$	`[9,14]`
6	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	`[10,13]`
7	$\left[\frac{3}{4}, -x, \frac{3}{4}\right]$	`[11,15]`
8	$\left[\frac{3}{4}, x+\frac{1}{2}, \frac{3}{4}\right]$	`[12,16]`

Table 10: Wyckoff site: `8j`, site symmetry: `.2.`

No.	position	mapping
1	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	`[1,6]`
2	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{1}{4}\right]$	`[2,5]`
3	$\left[\frac{1}{4}, x, \frac{3}{4}\right]$	`[3,7]`
4	$\left[\frac{1}{4}, \frac{1}{2}-x, \frac{3}{4}\right]$	`[4,8]`

continued ...

Table 10

No.	position	mapping
5	$\left[-x,\ \frac{3}{4},\ \frac{3}{4}\right]$	[9,14]
6	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{3}{4}\right]$	[10,13]
7	$\left[\frac{3}{4},\ -x,\ \frac{1}{4}\right]$	[11,15]
8	$\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[12,16]

Table 11: Wyckoff site: 8k, site symmetry: ..2

No.	position	mapping
1	$[x,\ x,\ 0]$	[1,7]
2	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ 0\right]$	[2,8]
3	$\left[\frac{1}{2}-x,\ x,\ \frac{1}{2}\right]$	[3,5]
4	$\left[x,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[4,6]
5	$[-x,\ -x,\ 0]$	[9,15]
6	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$	[10,16]
7	$\left[x+\frac{1}{2},\ -x,\ \frac{1}{2}\right]$	[11,13]
8	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[12,14]

Table 12: Wyckoff site: 8l, site symmetry: ..2

No.	position	mapping
1	$\left[x,\ x,\ \frac{1}{2}\right]$	[1,7]
2	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[2,8]
3	$\left[\frac{1}{2}-x,\ x,\ 0\right]$	[3,5]
4	$\left[x,\ \frac{1}{2}-x,\ 0\right]$	[4,6]
5	$\left[-x,\ -x,\ \frac{1}{2}\right]$	[9,15]
6	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[10,16]
7	$\left[x+\frac{1}{2},\ -x,\ 0\right]$	[11,13]
8	$\left[-x,\ x+\frac{1}{2},\ 0\right]$	[12,14]

Table 13: Wyckoff site: 8m, site symmetry: ..m

No.	position	mapping
1	$[x,\ -x,\ z]$	[1,15]
2	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ z\right]$	[2,16]
3	$\left[x+\frac{1}{2},\ x,\ z+\frac{1}{2}\right]$	[3,13]
4	$\left[-x,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[4,14]
5	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{2}-z\right]$	[5,11]
6	$\left[x,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[6,12]
7	$[-x,\ x,\ -z]$	[7,9]
8	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ -z\right]$	[8,10]

Table 14: Wyckoff site: `16n`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	`[2]`
3	$\left[\frac{1}{2}-y,\ x,\ z+\frac{1}{2}\right]$	`[3]`
4	$\left[y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[4]`
5	$\left[\frac{1}{2}-x,\ y,\ \frac{1}{2}-z\right]$	`[5]`
6	$\left[x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	`[6]`
7	$[y,\ x,\ -z]$	`[7]`
8	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ -z\right]$	`[8]`
9	$[-x,\ -y,\ -z]$	`[9]`
10	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$	`[10]`
11	$\left[y+\frac{1}{2},\ -x,\ \frac{1}{2}-z\right]$	`[11]`
12	$\left[-y,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[12]`
13	$\left[x+\frac{1}{2},\ -y,\ z+\frac{1}{2}\right]$	`[13]`
14	$\left[-x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[14]`
15	$[-y,\ -x,\ z]$	`[15]`
16	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z\right]$	`[16]`